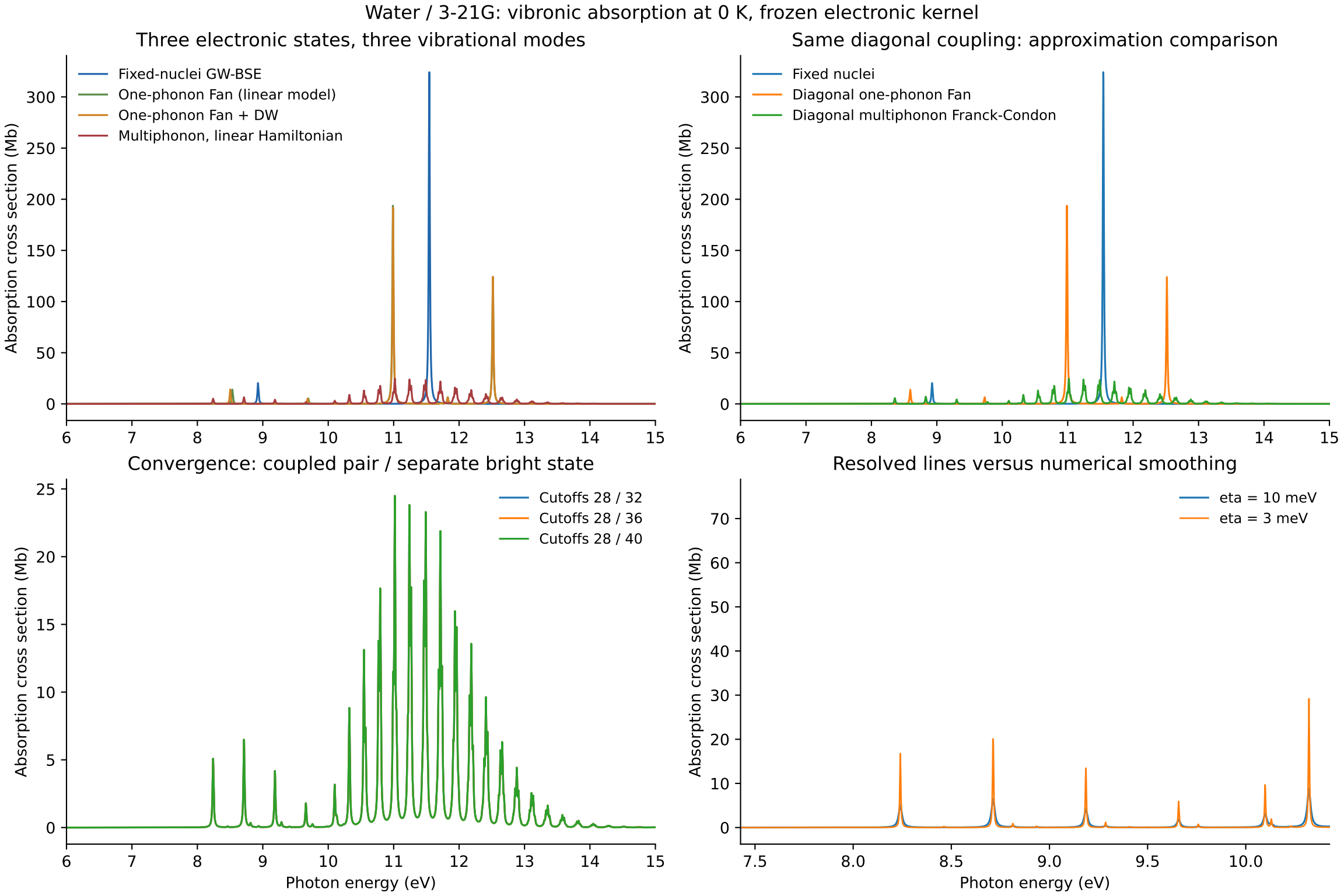
Water / 3-21G: vibronic absorption at 0 K, frozen electronic kernel
Three electronic states, three vibrational modes
Fixed-nuclei GW-BSE
One-phonon Fan (linear model)
One-phonon Fan + DW
Multiphonon, linear Hamiltonian
Absorption cross section (Mb)
Same diagonal coupling: approximation comparison
Fixed nuclei
Diagonal one-phonon Fan
Diagonal multiphonon Franck-Condon
Absorption cross section (Mb)
Convergence: coupled pair / separate bright state
Cutoffs 28 / 32
Cutoffs 28 / 36
Cutoffs 28 / 40
Absorption cross section (Mb)
Photon energy (eV)
Resolved lines versus numerical smoothing
eta = 10 meV
eta = 3 meV
Absorption cross section (Mb)
Photon energy (eV)